

RF Signal Detector



PETRE TZV. PETROV

This simple circuit can be used to trace the presence of RF signals and electromagnetic noise in your residential area, office or shop. It can be a useful tool while testing or designing RF circuits. It can also be used to detect electrical noise in your premises.

Electromagnetic noises can be produced by sparking in electrical in-

stallations or other sources. To detect the presence of RF signal or noise, you need a wide-band receiver that can capture these signals.

Circuit and working

Fig. 1 shows the circuit of RF signal detector. It consists of a telescopic antenna, input protection resistor (R1), two diodes (D1 and D2), selector switch (S1), preamplifier with a low-noise and high-gain transistor (T1), and audio amplifier IC LM386 (IC1).

Switch S1 is used to select between high and low sensitivities of the circuit. When high sensitivity is selected, gain of the transistor stage with T1 is added to gain of IC LM386. When low

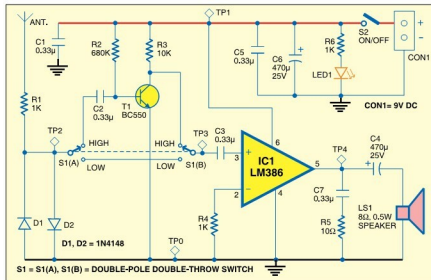


Fig. 1: Circuit of RF signal detector

PARTS LIST

Semiconductors:

- IC1 - LM386 low-power audio amplifier
- T1 - BC550 npn
- LED1 - 5mm red LED
- D1, D2 - 1N4148

Resistors (all ¼-watt, 45% carbon):

- R1, R4, R6 - 1-kilo-ohm
- R2 - 680-kilo-ohm
- R3 - 10-kilo-ohm
- R5 - 10-ohm

Capacitors:

- C1-C3, C5, C7 - 0.33μF ceramic disk
- C4, C6 - 470μF, 25V electrolytic

Miscellaneous:

- Ant - 10cm wire
- S1 - DPDT switch
- S2 - On/off switch
- LS1 - 8-ohm, 0.5W speaker
- CON1 - Two-pin connector

Test Points

Test point	Details
TP0	GND
TP1	9V
TP2	Input signal (noise picked up by antenna)
TP3	Amplified signal in case of high sensitivity selection
TP4	Output amplified signal

sensitivity is selected, gain of only IC LM386 is used, which is around 20. In order to keep the circuit simple and compact, no volume control is included.

When there is a strong electromagnetic signal near the antenna, e.g., from a mobile phone, telephone or electrical motor, you can hear audible sound from the speaker (LS1). As you bring this detector closer to the RF or noise-transmitting source, the sound from the speaker becomes louder. Thus you can trace the exact location of the

**Calling Upon
All Electronics Manufacturers,
and Channel Partners
On Facebook...**

Join **Electronics Bazaar facebook** at
<http://www.facebook.com/ElectronicsB2B>

and share your views on Electronics B2B issues.
Also, GET UPDATES FROM THE ELECTRONICS B2B INDUSTRY

